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SI 305

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Final Project Retrospective Commentary

I really enjoyed working on this project, not only because it allowed me to work on a public health topic that is extremely relevant in my hometown, but also because it taught me about what actual data analysis looks like.

The first challenge I faced with this project came from my first research question, which showed that carcinogen releases had essentially no correlation with cancer incidence rates at a county level. This made me assume that the analysis was a complete failure, as I had expected to find a clear positive correlation between total carcinogen release and cancer incidence. That being said, this part of the analysis showed that obesity, smoking, and poverty were strong predictors of cancer incidence, helping shape my second research question and create a strong and targeted recommendation for promoting public health. Moving forward, I will be more patient with unexpected results, as this project proved that no correlation between two variables can be incredibly meaningful.

The second challenge I encountered came from the data preparation rather than the actual analysis. I spent hours looking for relevant datasets, eventually finding the Michigan Cancer Atlas, which provided incredibly helpful data. However, the atlas only had smoking and obesity rates at a prosperity region level, not a county level. This required me to map each county to their respective prosperity levels, limiting analysis accuracy. For future projects, I will prioritize finding datasets from one of the sources discussed in class.